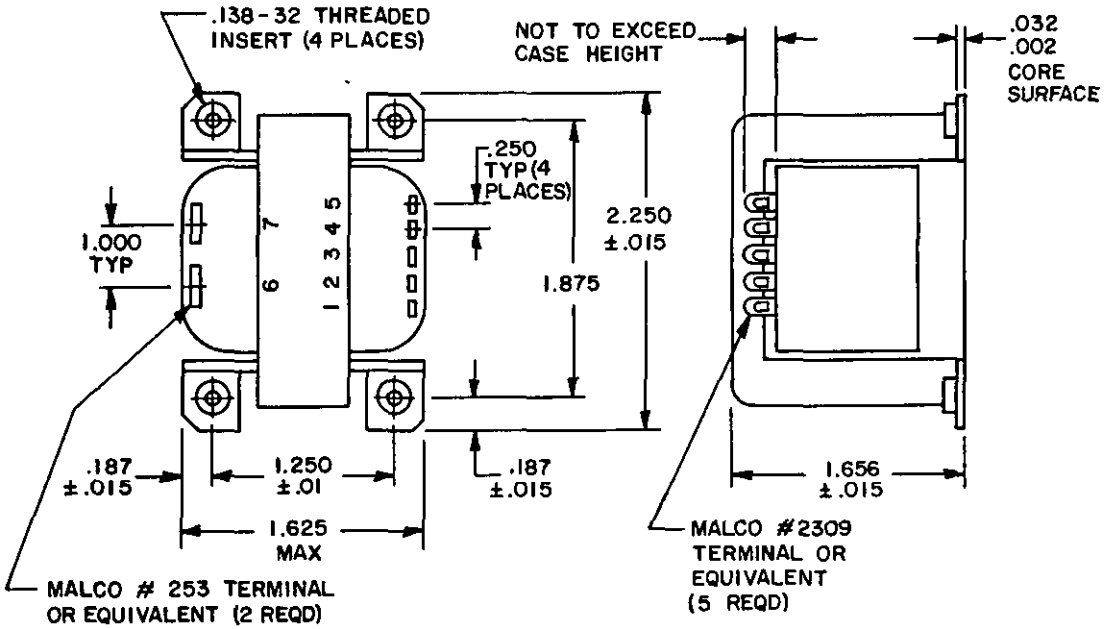
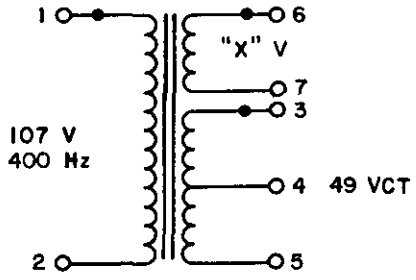
MILITARY SPECIFICATION SHEET

TRANSFORMERS, POWER, STEP-DOWN, TF5T03ZZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.





WORKING VOLTAGE: 120 Vrms
ALTITUDE: 70 k FT MAX
(SEE TABLE I FOR VALUE "X")

INCHES	MM
.002	.05
.015	.38
.032	.81
.138	3.51
.187	4.75
.250	6.35
1.000	25.40
1.250	31.75
1.625	41.28
1.656	42.06
1.875	47.62
2.250	57.15

CIRCUIT DIAGRAM AND MARKING

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on side of case.
4. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

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FSC 5950

REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.
Primary voltage (1-2): 107 volts rms, 400 hertz ±20.
Volt-ampere rating: 100 volt-amperes.
Primary current (1-2): 1.0 ampere.
Working voltage: 120 volts rms.

TABLE I. Electrical ratings.

Dash no.	Secondary voltage (V ±10%)		Secondary current (A ±10%)	
	(3-5)	(6-7)	(3-5)	(6-7)
01	49 ct	6.35	0.010	15.9
02	49 ct	6.56	0.010	15.3

Design and construction:
Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Encapsulated.
Terminals: Solder terminal, 1 and 2 shall be Malco No. 253 or equivalent and 3, 4, 5, 6 and 7 shall be Malco No. 2309 or equivalent.
Terminal height: Not to exceed case height.
Weight: 9.702 ounces, maximum.
Altitude: 70,000 feet maximum.
Operating temperature range: -55°C to +170°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage:
At sea level: 1,000 volts rms.
At reduced barometric pressure: 300 volts rms.

Electrical characteristics: See table II.
Rated load: With 107 volts rms, 400 hertz across (1-2), the secondary voltages shall be as specified in table I.
Polarity: Additive, with terminals 2 and 3, and 5 and 6 connected.

TABLE II. Electrical characteristics.

Dash no.	No load: With 107 Vac and 400 Hz in (1-2)				DC resistance (Ω max)		
	Current mA, (max)	Power W (max)	Voltage V ±10%	Voltage V ±10%	(1-2)	(3-5)	(6-7)
01	(1-2) 250	(1-2) 30	(3-5) 49 ct	(6-7) 6.35	3.0	20	0.0118
02	250	30	49 ct	6.56	3.0	20	0.0110

Temperature rise: 99°C with 107 volts rms, 400 hertz across (1-2) at an ambient temperature of 71°C.

Marking location: See figure 1.

Part number: M27/220- (dash number from table I).

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, WC, ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

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